

Smart Home-Hospitalization: Balancing Technology and Care

Solving hospital overcrowding by empowering caregivers with intelligent IoT.



Italy's Demographic Inversion Demands a New Care Model

The Capacity Crunch

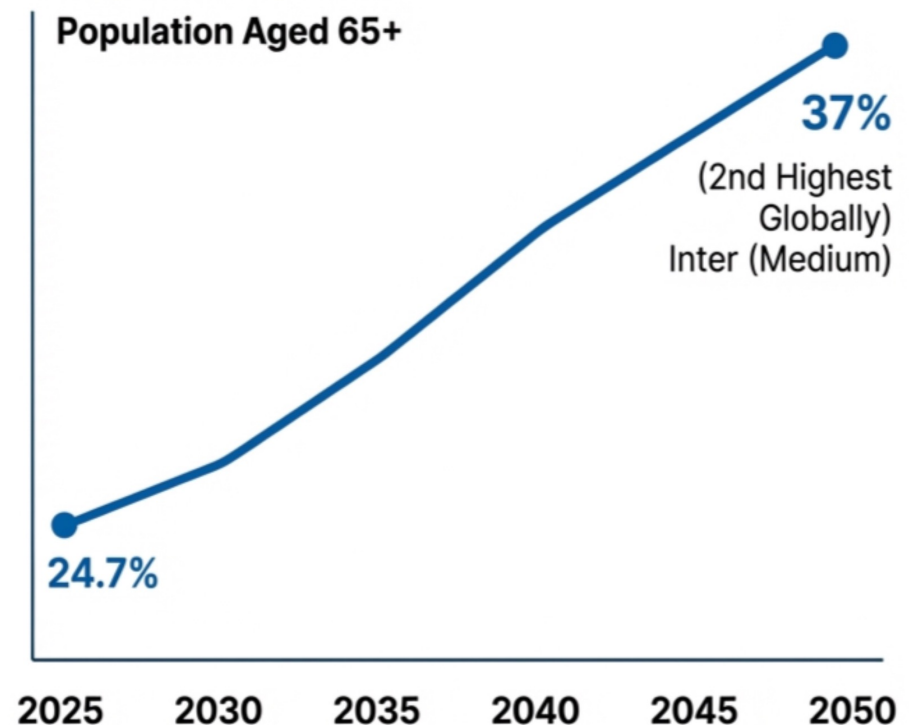
With 15.9% of the population reporting difficulties with complex instrumental activities, demand for beds drastically outstrips hospital supply.

Fragmented System

Current services are strictly divided between health and social care, leading to uneven access and fragmented outcomes.

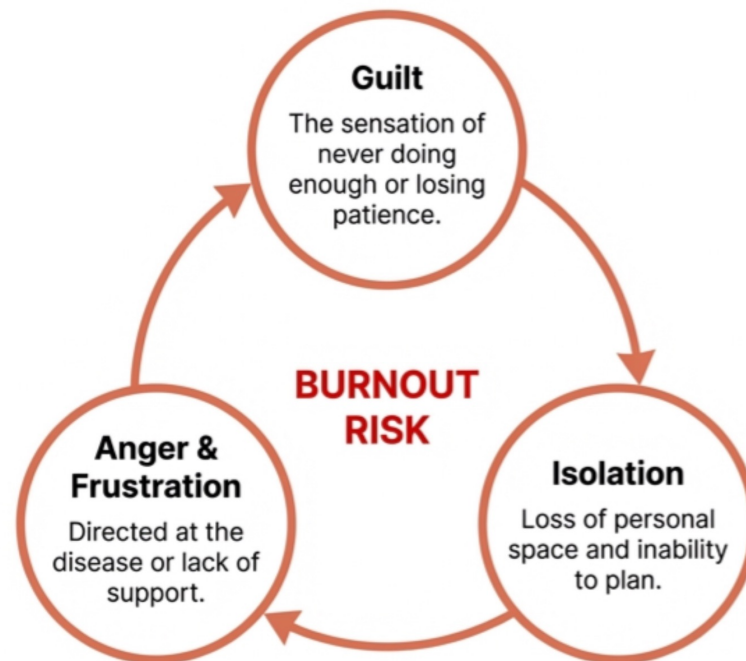
The Conclusion

Home hospitalization is no longer a preference; it is the only viable systemic solution to manage the surge in chronic conditions.

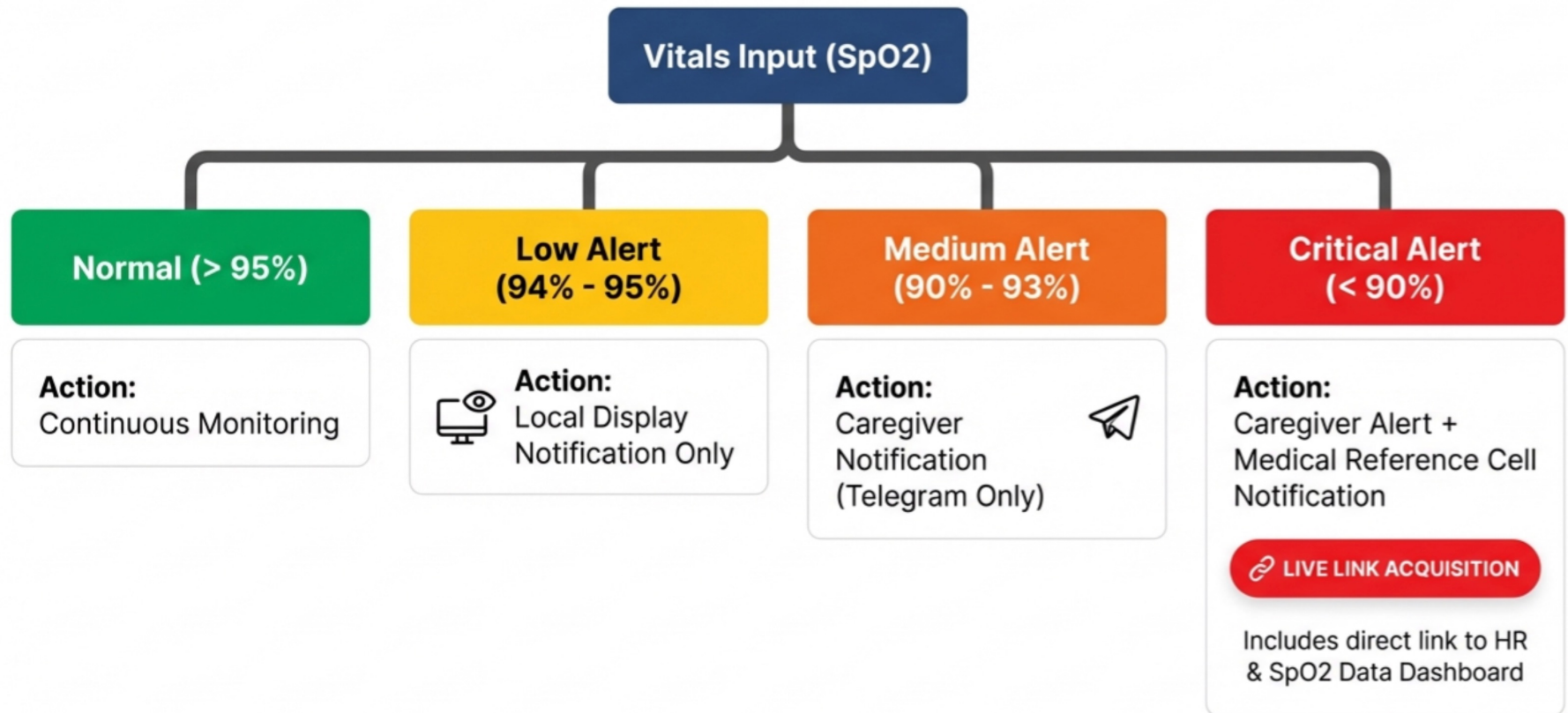


The Invisible Crisis: The Caregiver Burden

Caregiving is a constant physical and emotional commitment, often without pause or recognition.



Vitals Input (SpO2)



The IoT Solution: Real-Time Monitoring & Safety

3. OLED Feedback for Local Reassurance

Triggers a Direct Alert to caregiver and dashboard. emtance (118) to prevent false alarms.



4. SOS Button

Triggers a Direct Alert to the caregiver and dashboard. Does NOT immediately contact (118) to prevent false alarms.

1. Particle IoT Board (Connectivity & Processing)

2. SpO2 & Heart Rate Sensors



Bridging the Gap Between Home and Hospital



Remote Review

Medical professionals access the Live Dashboard link generated during critical alerts to view trends, not just snapshots.

Efficiency

Eliminates manual vitals recording, reducing caregiver workload and error margins.

The 3 Pillars of Caregiver Relief

Peace of mind is the ultimate deliverable.



Automation

By automating “constant manual observation,” the device acts as a second pair of eyes, removing the need for constant vigilance.



Reassurance

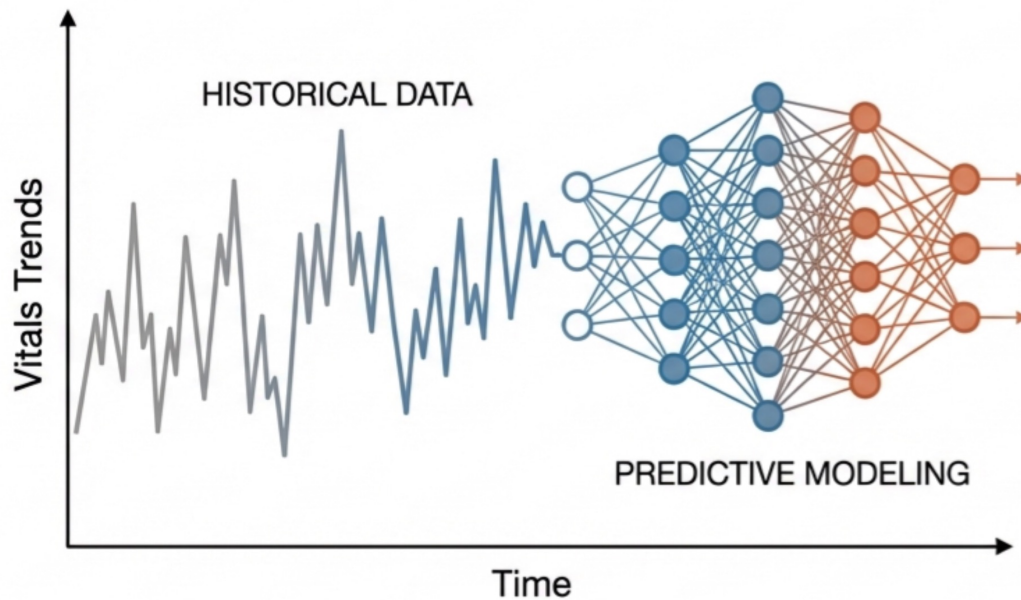
The caregiver knows the system is watching, reducing the deep-seated fear of missing a critical event



Freedom

Allows the caregiver to sleep or take a break with the security that a Critical Alert will wake them if necessary.

Future Perspectives: From Reaction to Prediction



- **Next-Gen Implementation: Machine Learning**
Utilizing historical data to analyze SpO2 and Heart Rate trends over time.
- **Enhanced Triage**
Using ML to predict health declines before they hit the Critical Alert threshold ($< 90\%$).

A scalable, data-driven model to support Italy's aging population, transforming home care from a burden into a sustainable, safe reality.